

RL Homework 2: DDPG, TRPO, and PPO

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April 25, 2025

Problem 1 (Surrogate Function in TRPO)

Property (i): $L_{\pi_{\theta_1}}(\pi_{\theta_1}) = \eta(\pi_{\theta_1})$

Proof. By definition, we have

$$L_{\pi_{\theta_1}}(\pi_{\theta_1}) = \eta(\pi_{\theta_1}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \pi_{\theta_1}(a | s) A^{\pi_{\theta_1}}(s, a) \quad (1)$$

$$= \eta(\pi_{\theta_1}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \pi_{\theta_1}(a | s) (Q^{\pi_{\theta_1}}(s, a) - V^{\pi_{\theta_1}}(s)) \quad (2)$$

$$= \eta(\pi_{\theta_1}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \left(\sum_{a \in \mathcal{A}} \pi_{\theta_1}(a | s) Q^{\pi_{\theta_1}}(s, a) - \sum_{a \in \mathcal{A}} \pi_{\theta_1}(a | s) V^{\pi_{\theta_1}}(s) \right) \quad (3)$$

$$= \eta(\pi_{\theta_1}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \left(V^{\pi_{\theta_1}}(s) - V^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \pi_{\theta_1}(a | s) \right) \quad (4)$$

$$= \eta(\pi_{\theta_1}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) (V^{\pi_{\theta_1}}(s) - V^{\pi_{\theta_1}}(s) \cdot 1) = \eta(\pi_{\theta_1}) \quad (5)$$

□

Property (ii): $\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) = \nabla_{\theta} \eta(\pi_{\theta})$

Proof. Note that

$$\nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) = \nabla_{\theta} \eta(\pi_{\theta_1}) + \nabla_{\theta} \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \pi_{\theta}(a | s) A^{\pi_{\theta_1}}(s, a) \quad (6)$$

$$= \nabla_{\theta} \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \pi_{\theta}(a | s) A^{\pi_{\theta_1}}(s, a) \quad (7)$$

$$= \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) A^{\pi_{\theta_1}}(s, a) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) \quad (8)$$

since ∇_{θ} does not rely on a fixed θ_1 and $\nabla_{\theta} \eta(\pi_{\theta_1}) = 0$.

Also, we have

$$\nabla_{\theta} \eta(\pi_{\theta}) = \nabla_{\theta} V^{\pi_{\theta}}(\mu) = \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta}}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \quad (9)$$

from the policy gradient theorem. When $\theta = \theta_1$, $A^{\pi_{\theta_1}}(s, a) = Q^{\pi_{\theta_1}}(s, a) - V^{\pi_{\theta_1}}(s)$,

$$\nabla_{\theta} \eta(\pi_{\theta}) = \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta}}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi_{\theta}}(s, a) \quad (10)$$

$$= \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) Q^{\pi_{\theta_1}}(s, a) \quad (11)$$

$$= \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) (A^{\pi_{\theta_1}}(s, a) + V^{\pi_{\theta_1}}(s)) \quad (12)$$

$$= \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) A^{\pi_{\theta_1}}(s, a) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) V^{\pi_{\theta_1}}(s) \sum_{a \in \mathcal{A}} \nabla_{\theta} \pi_{\theta}(a | s) \quad (13)$$

$$= \nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) V^{\pi_{\theta_1}}(s) \nabla_{\theta} \sum_{a \in \mathcal{A}} \pi_{\theta}(a | s) \quad (14)$$

$$= \nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) V^{\pi_{\theta_1}}(s) \cdot (\nabla_{\theta} 1) \quad (15)$$

$$= \nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}) + \sum_{s \in \mathcal{S}} d_{\mu}^{\pi_{\theta_1}}(s) V^{\pi_{\theta_1}}(s) \cdot 0 = \nabla_{\theta} L_{\pi_{\theta_1}}(\pi_{\theta}). \quad (16)$$

□

Problem 2 (Solving TRPO Under Approximation Using Duality)

(a) Dual Function $D(\lambda)$ and the Optimal λ^*

Proof. Since $\mathcal{L}(\theta, \lambda)$ is a quadratic function in θ , thus convex in θ , the minimizer θ^* appears when $\nabla_{\theta}\mathcal{L}(\theta, \lambda)|_{\theta=\theta^*} = 0$. Note that $\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k}$ is a constant vector, and we have

$$\nabla_{\theta}\mathcal{L}(\theta, \lambda) = -\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k} + \lambda H_{\theta_k}(\theta - \theta_k) \quad (17)$$

where $\nabla_{\theta}(\theta - \theta_k)^{\top}(\theta - \theta_k) = 2(\theta - \theta_k)$. Plugging θ^* , we have

$$\nabla_{\theta}\mathcal{L}(\theta^*, \lambda) = -\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k} + \lambda H_{\theta_k}(\theta^* - \theta_k) = 0, \quad (18)$$

and

$$\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k} = \lambda H_{\theta_k}(\theta^* - \theta_k), \quad \theta^* - \theta_k = \frac{1}{\lambda} H_{\theta_k}^{-1}(\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k}). \quad (19)$$

Thus,

$$D(\lambda) = \min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta, \lambda) = \mathcal{L}(\theta^*, \lambda) \quad (20)$$

$$= -(\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top}(\theta^* - \theta_k) + \lambda \left(\frac{1}{2}(\theta^* - \theta_k)^{\top} H_{\theta_k}(\theta^* - \theta_k) - \delta \right) \quad (21)$$

$$= -(\lambda H_{\theta_k}(\theta^* - \theta_k))^{\top}(\theta^* - \theta_k) + \lambda \left(\frac{1}{2}(\theta^* - \theta_k)^{\top} H_{\theta_k}(\theta^* - \theta_k) - \delta \right) \quad (22)$$

$$= \frac{-1}{2\lambda}(\theta^* - \theta_k)^{\top} H_{\theta_k}(\theta^* - \theta_k) - \lambda\delta \quad (23)$$

$$= \frac{-1}{2\lambda} \left((\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1}(\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right) - \lambda\delta. \quad (24)$$

Since $D(\lambda)$ is a concave function for $\lambda > 0$, the maximizer λ^* appears when $\frac{d}{d\lambda}D(\lambda)|_{\lambda=\lambda^*} = 0$. Thus,

$$\frac{d}{d\lambda}D(\lambda^*) = \frac{d}{d\lambda} \left(\frac{-1}{2\lambda^*} \left((\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1}(\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right) - \lambda^*\delta \right) \quad (25)$$

$$= \frac{1}{2\lambda^{*2}} \left((\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1}(\nabla_{\theta}L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right) - \delta = 0, \quad (26)$$

then

$$\lambda^{*2} = \frac{1}{2\delta} \left((\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right), \quad (27)$$

$$\lambda^* = \sqrt{\frac{1}{2\delta} \left((\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right)} \geq 0. \quad (28)$$

□

(b) Primal Solution θ^* and Step Size α

Proof. Consider our primal constraint,

$$\frac{1}{2}(\theta - \theta_k)^{\top} H_{\theta_k}(\theta - \theta_k) - \delta \leq 0, \quad \frac{1}{2}(\theta - \theta_k)^{\top} H_{\theta_k}(\theta - \theta_k) \leq \delta, \quad (29)$$

which is a quadratic function in θ . If $\delta < 0$, then there won't exist any feasible solution for TRPO updating. If $\delta = 0$, then only the trivial solution $\theta = \theta_k$ is feasible, also impractical for TRPO updating. Thus, we have $\delta > 0$. Consider a feasible solution $\theta = \theta_k$, we have

$$\frac{1}{2}(\theta - \theta_k)^{\top} H_{\theta_k}(\theta - \theta_k) - \delta = \frac{1}{2}(\theta_k - \theta_k)^{\top} H_{\theta_k}(\theta_k - \theta_k) - \delta = -\delta < 0, \quad (30)$$

which indicates that such feasible solution is strictly feasible. Since we have the primal optimization problem being convex and there also exists a strictly feasible solution $\theta = \theta_k$, by Slater's Condition, the strong duality holds. This implies that θ^* is the minimizer of $\mathcal{L}(\theta, \lambda)$ if and only if λ^* is the maximizer of $D(\lambda)$. Thus, we have

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta, \lambda) = \arg \min_{\theta \in \mathbb{R}^d} \mathcal{L}(\theta, \lambda^*) \quad (31)$$

And following Equation (19), which we derived from the optimality of θ^* , we have

$$\theta^* - \theta_k = \frac{1}{\lambda} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) = \frac{1}{\lambda^*} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}), \quad (32)$$

$$\theta^* = \theta_k + \frac{1}{\lambda^*} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) = \theta_k + \alpha H_{\theta_k}^{-1} \nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}, \quad (33)$$

where

$$\alpha = \frac{1}{\lambda^*} = \left(\frac{1}{2\delta} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k})^{\top} H_{\theta_k}^{-1} (\nabla_{\theta} L_{\theta_k}(\theta)|_{\theta=\theta_k}) \right)^{-\frac{1}{2}} > 0 \quad (34)$$

given H is positive definite and $\delta > 0$.

□

Problem 3 (PPO With a Clipped Objective)

The only differences appear in regions 3 and 6 of Figure 1, where the curve for $\tilde{L}_{s,a}^{\text{clip}}(\theta; \theta_k)$ is drawn as a horizontal line instead of following the diagonal.

1. Case 1: $A^{\pi_{\theta_k}}(s, a) > 0$.

(a) The unclipped surrogate is

$$L_{s,a}^{\text{clip}}(\theta; \theta_k) = \frac{\pi_{\theta}(a | s)}{\pi_{\theta_k}(a | s)} A^{\pi_{\theta_k}}(s, a).$$

(b) The clipped surrogate takes the lower bound

$$\tilde{L}_{s,a}^{\text{clip}}(\theta; \theta_k) = (1 - \epsilon) A^{\pi_{\theta_k}}(s, a).$$

(c) This corresponds to the regime

$$\frac{\pi_{\theta}(a | s)}{\pi_{\theta_k}(a | s)} \leq 1 - \epsilon.$$

2. Case 2: $A^{\pi_{\theta_k}}(s, a) < 0$.

(a) The unclipped surrogate remains

$$L_{s,a}^{\text{clip}}(\theta; \theta_k) = \frac{\pi_{\theta}(a | s)}{\pi_{\theta_k}(a | s)} A^{\pi_{\theta_k}}(s, a).$$

(b) The clipped surrogate takes the upper bound

$$\tilde{L}_{s,a}^{\text{clip}}(\theta; \theta_k) = (1 + \epsilon) A^{\pi_{\theta_k}}(s, a).$$

(c) This corresponds to the regime

$$\frac{\pi_{\theta}(a | s)}{\pi_{\theta_k}(a | s)} \geq 1 + \epsilon.$$

Thus, the only distinction arises for very small importance-weighted advantages. L^{clip} preserves these small advantages as long as they satisfy its one-sided clipping bound, whereas $\tilde{L}^{\text{clip}}$ employs symmetric two-sided clipping, fully discarding any advantages that fall outside the prescribed interval.

Problem 4 (Deep Deterministic Policy Gradient for Continuous Control)

(a) Pendulum-v0 with DDPG

Our network structure follows:

1. Actor, three fully connected layers with GELU activations and one fully connected output layer with *tanh* activation.
2. Critic, three fully connected layers, where the first layer also took the action as the input, with RELU activations, and one fully connected output layer.
3. The hidden size of both network is 128.

Our hyperparameters follow:

1. $\gamma = 0.98$
2. $\tau = 0.001$
3. OU noise scale, $\text{noise_scale} = 0.05$
4. Actor learning rate, $\text{lr_a} = 0.0001$
5. Critic learning rate, $\text{lr_c} = 0.00059999999999999997$

We use the batch size of 64, and warm up for 3000 environment steps. We use the Bayesian Optimization among 100 different combinations of hyperparameters but still could NOT solve the Pendulum-v0 environment even in 1000 episodes (Fig. 1).

We also record the gradient norm ratio between Actor and Critic. Low ratio indicated that the Actor could not obtain a good enough training signal from Critic. The training dynamic suggest that our Critic is overfitting the earlier noise. This is evidenced by the dropping of Critic's loss to nearly zero in the early steps, and the low gradient norm ratio between Actor and Critic (Fig. 2).

(a) HalfCheetah-v3 with DDPG

Our HalfCheetah environment performs even worse, please refer to the training results (Fig. 3 and Fig. 4). The network structure are similar, but with additional layer normalization and warm up for 10000 steps.

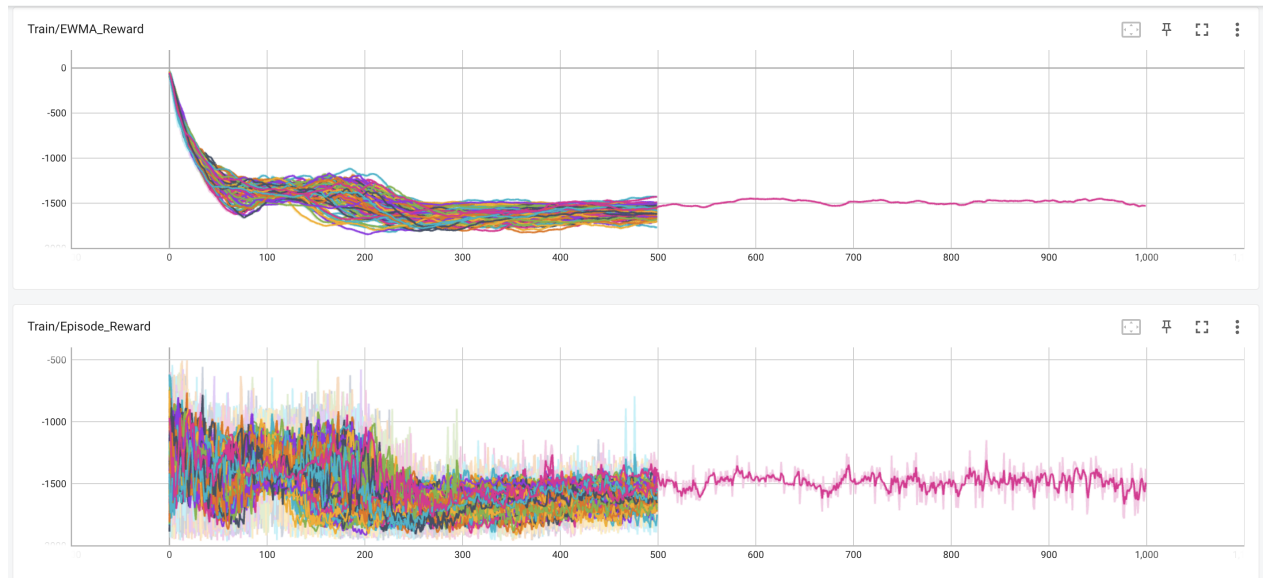


Figure 1: Pendulum Train

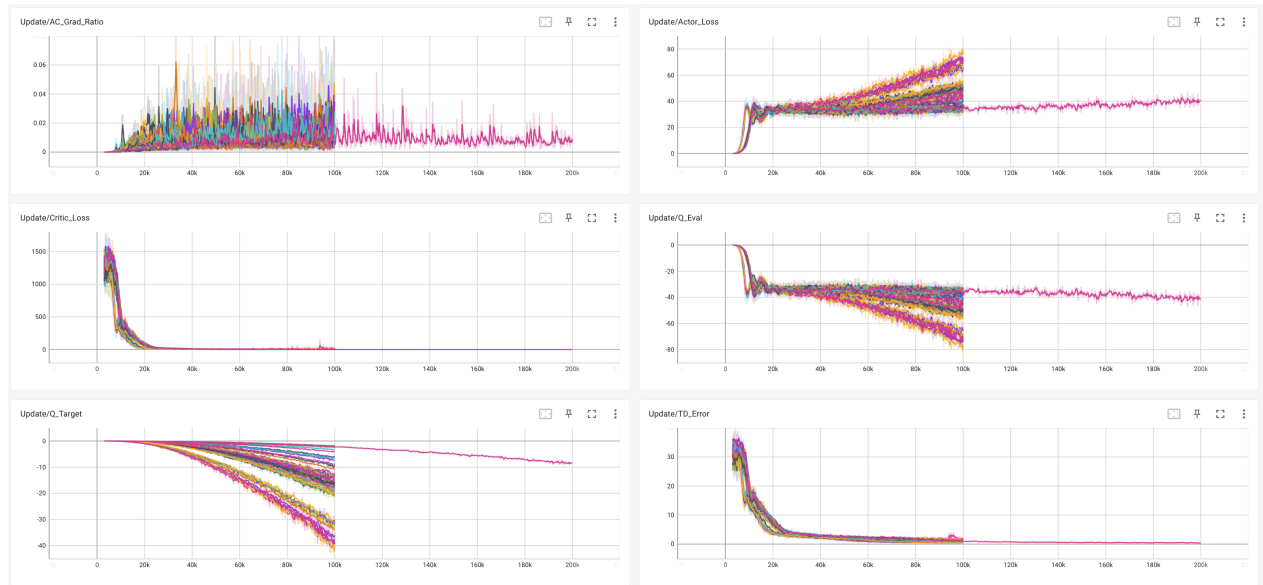


Figure 2: Pendulum Update

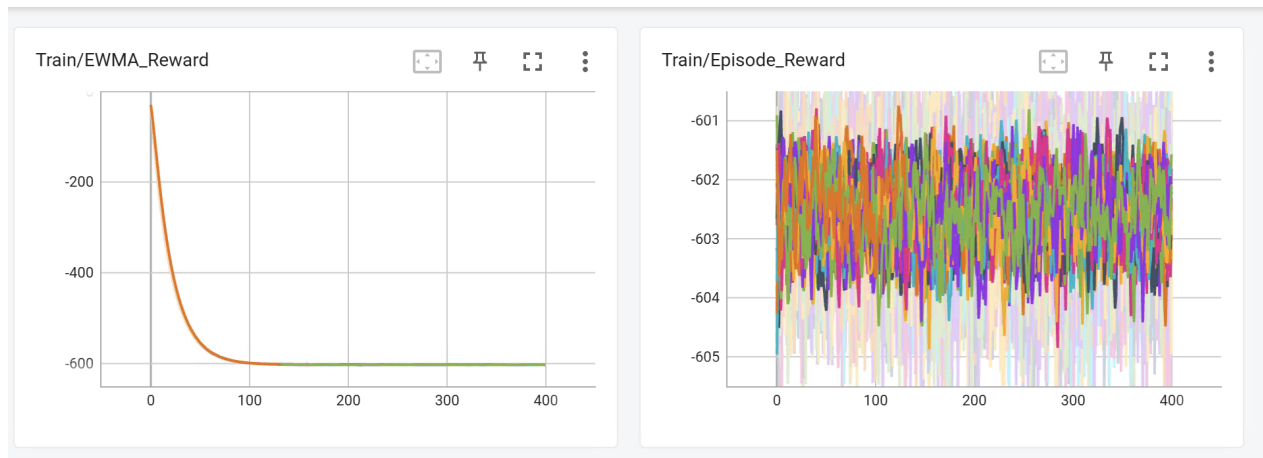


Figure 3: HalfCheetah Train

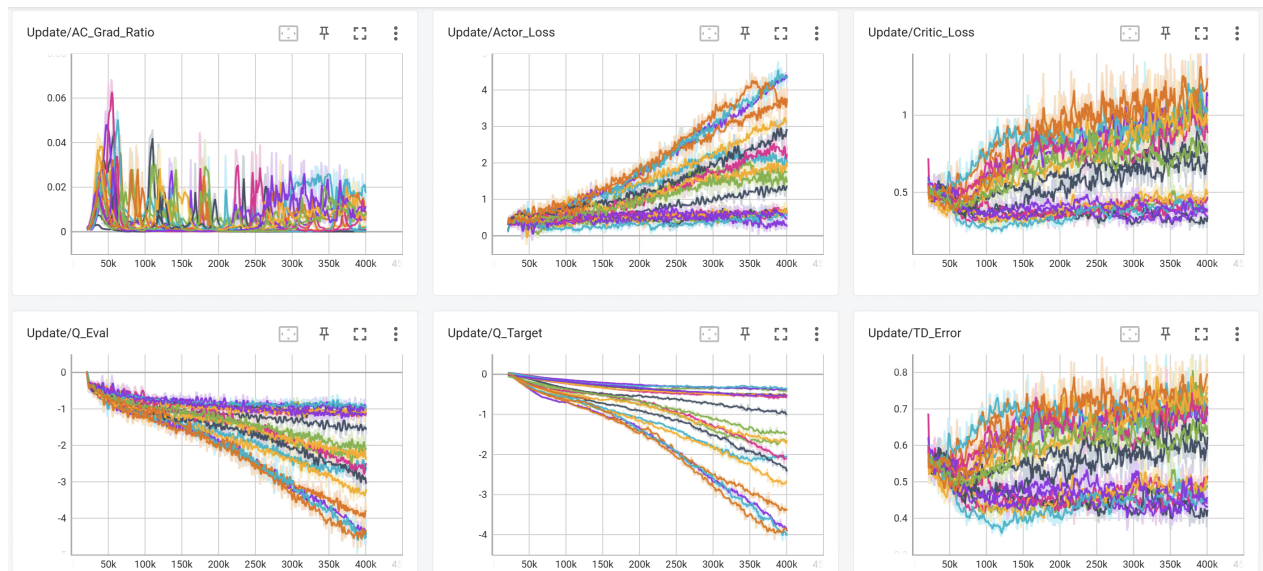


Figure 4: HalfCheetah Update